⛔ **GATED · 진입 보류**

한국인 Graves' 병 HLA susceptibility

A Pan-Asian autoimmune-thyroid HLA landscape — gated until 4 entry conditions are met

> *Bundang SNUH Graves' cohort + Chu X et al. 2018 Han Chinese GD (n=1,468) Pan-Asian random-effects forest meta + cookHLA SNP-based 4-digit imputation cross-platform validation.* ***2026-05-04 기준 status****:* ***GATED*** *— Paper 1 bioRxiv 제출 (target 2026-06-13) + Paper 2 Task A/B/C 완료 + Bundang prospective response (n>50) + Yu professor trajectory 합의 4 조건 모두 충족 전 진입 X. 본 briefing 은 entry 후 진행될 5-pillar plan 의 advance preview.*

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TARGET VENUE · J AUTOIMMUNITY REACH → FRONT IMMUNOL SAFE → HLA FALLBACK DATE · 2026-05-04 · GATED

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§ 1

GATED status — entry blocked

2026-05-04 기준 4 entry conditions 0/4 met. Source file touch (read 포함) 금지.

> ⛔ 진입 보류 — ADVISOR 2026-05-04 DIRECTIVE
>
> **Paper 3 = Korean Graves' disease HLA susceptibility, Pan-Asian autoimmune-thyroid landscape.** Paper 1 (cancer) / Paper 2 (autoimmune-overlap PTC) 와 완전 isolate. 4 entry conditions 모두 충족 전 진입 X. 미충족 시 Paper 3 file touch 금지 (source file read 도 금지). Marathon mode 5/4–6/13 은 Paper 1 bioRxiv + Paper 2 manuscript 전용 — Paper 3 는 그 이후.

Why gated. Paper 1 · 2 · 3 cross-contamination (HT term in GD paper, GD term in HT paper, PTC mechanism in autoimmune paper) 시 reviewer 가 scope confusion 으로 reject. 본 paper 의 disease 는 **"갑상선을 자극하는 자가항체가 생성되어 갑상선 호르몬이 과다 분비되는 자가면역 질환"** (advisor 인용; Graves' =

TSAb-mediated hyperthyroidism). Paper 2 의 destructive lymphocytic infiltration 와 mechanism 정반대 — disease specificity (GD ≠ HT mechanism) 가 distinct paper 정당화의 핵심.

본인 unique angle. **cookHLA** (Nat Commun first-author HLA imputation tool) + Pan-Asian extension. 두 분 (주영석 / 최정균) 도 갖고 있지 않은 본인 고유 자산. SNP-based 4-digit HLA imputation 의 cross-platform validation 이 reviewer 의 "RNA-seq 기반 imputation 만으로 충분?" 우려 사전 차단.

§ 2

4 entry conditions ★

4/4 충족 시 진입 권한 획득. 3개 이하 시 file touch 금지.

1

Paper 1 bioRxiv 제출 완료. 5/4–6/13 marathon W1–W6 마지막 day 의 W6 cover letter + submission 완료 후 진입 권한 활성화. Paper 1 이 아직 W0 prep day 단계.

PENDING

target 2026-06-13

2

Paper 2 Task A / B / C 완료. Pillar 1 STRONG 격상 + 4-cohort meta polish + figure final SVG. 현재 advisor discussion brief 완료, 본 작업 진행 중.

PENDING

target 2026-06-15

3

Bundang SNUH Graves' cohort 응답 (n > 50 STRONG GO 또는 6주 무응답 freeze). 분당서울대 prospective Graves' cohort 의 outreach 결과 대기. 응답 시 RNA-seq + DNA panel 조건 협상.

EXTERNAL

분당 응답 대기

4

Yu professor Paper 3 trajectory 합의. Paper 1 + 2 완료 후 Paper 3 venue ladder (J Autoimmunity reach vs Front Immunol safe) 결정 + Pan-Asian framing 승인.

PENDING

Paper 1 · 2 후 미팅

CURRENT STATE — 0 / 4 MET

Paper 3 file touch 권한 0%. 본 status report 는 memory `v19_paper3_GD_gating.md` 와 `v17_graves_pivot.md` 만 참조 — Paper 3 source file (project/results/paper3_*) 미접근. Entry 후 Task 1 부터 file generation 시작.

§ 3

Scope = autoimmune-thyroid only

Disease 정의 + 금지어 list (Paper 1 · Paper 2 territory term 등장 시 즉시 STOP + 재작성).

3.1 Disease definition (advisor 인용)

"갑상선을 자극하는 자가항체가 생성되어 갑상선 호르몬이 과다 분비되는 자가면역 질환" — 갑상선 자극 자가항체 (TSAb / TRAb) 가 TSH receptor 에 결합하여 갑상선 호르몬 과다 분비를 유발하는 자가면역 hyperthyroidism. 임상적으로 hyperthyroidism + 안구 돌출 + 갑상선 미만성 종대 (diffuse goiter) 의 triad. Graves 안병증 (Graves' ophthalmopathy) 동반율 25–50%.

3.2 금지어 list (Paper 1 · Paper 2 territory)

Table 1 — Forbidden term boundaries (Paper 3 scope)

TERRITORY	FORBIDDEN TERMS	REASON
Paper 2 (autoimmune-overlap PTC)	Hashimoto · lymphocytic thyroiditis · TLS · tertiary lymphoid structure · AICDA · BCR clonality · PTC+HT · Hashimoto-overlap	Destructive infiltration mechanism — GD 의 TSAb-stimulating mechanism 과 정반대
Paper 1 (cancer)	BRAF V600E · RET fusion · DM1/DM2 cluster · 8-gene RAI panel mechanism · LIBRETTO-001	Thyroid neoplasia mechanism — autoimmune disease 와 directly

TERRITORY	FORBIDDEN TERMS	REASON
	· selpercatinib	unrelated
Cross-paper	autoimmune-PTC · PTC+GD overlap · thyroid neoplasia framing	"Autoimmune thyroid disease" 로 정정. Paper 3 = autoimmune ONLY, neoplasia 미언급

§ 4

5-Pillar plan ★

진입 후 4–6주 in 진행될 5 layer evidence chain — 모두 PLANNED status.

I

Korean GD HLA cohort

Bundang prospective n > 50 · arcasHLA RNA-seq imputation

PLANNED

II

Chu 2018 replication

Han Chinese GD n=1,468 · J Med Genet 55(10):685–692

PLANNED

III

Pan-Asian forest meta

Korean GD + Chu 2018 · DerSimonian-Laird random-effects

PLANNED

IV

cookHLA SNP validation

SNP-based 4-digit imputation · cross-platform RNA-seq vs SNP

PLANNED

V

GD-specific haplotype

DRB1-DQA1-DQB1 trio + DPB1 LD · GD ≠ HT haplotype dissection

PLANNED

5 pillars 가 Pan-Asian autoimmune-thyroid HLA susceptibility 를 5가지 직교 layer (population-specific frequency · independent-cohort replication · cross-population meta · cross-platform genotyping · disease-specific haplotype) 로 동시 충족. Pillar I (Korean prospective) 가 Bundang 응답 n 에 따라 venue ladder 결정 (§ 6).

§ 5

Tasks 1–4 post-entry

Entry 후 4–6주 작업 흐름. Task 4 (Outline) 본인 voice 영역 — Claude generate 금지.

#	TASK	OUTPUT PATH	ESTIMATED TIME	DEPENDENCY
1	Bundang Graves' arcasHLA RNA-seq imputation	`project/results/paper3_bundang_GD_HLA/`	3–5 days (Azure)	Bundang fastq 수령
2	Pan-Asian forest meta (Korean GD + Chu 2018 Chinese GD)	`project/results/paper3_panasian_forest/`	2 days	Task 1 결과 + Chu 2018 summary statistics

#	TASK	OUTPUT PATH	ESTIMATED TIME	DEPENDENCY
3	cookHLA SNP cross-platform validation	`project/results/paper3_cookHLA_validation/`	3 days	Bundang SNP panel (협상 시)
4 ★	Outline (본인 voice 키보드)	`project/manuscript_p3/02_outline.md`	2 days	Tasks 1–3 결과 · venue n 기반 결정

Task 4 시점에 venue commit — Bundang n 확정 + effect size 확인 후 Yu 미팅 안건으로 올림. **Venue commit ladder** 는 § 6 참조.

§ 6

Venue ladder · 3 scenarios

Bundang n 기반 시나리오 분기. Yu 미팅 직전 결정.

SCENARIO A

n > 50 · STRONG GO

VENUE	**J Autoimmunity** (IF 12–14) reach default
FRAME	Pan-Asian forest meta + cookHLA SNP cross-platform 가 reach 정당화
COVER LETTER	Single version
PROBABILITY	Bundang 응답 시 ~40%

SCENARIO B

n = 30–50 · BORDERLINE

VENUE	J Autoimmunity submit 시도 + **Front Immunol** (IF 5–7) parallel
FRAME	Pan-Asian forest meta primary, Korean cohort supplementary
COVER LETTER	2 versions prep (J Autoim + Front Immunol)
PROBABILITY	~35%

SCENARIO C

n < 30 또는 6주 freeze

VENUE	J Autoimmunity 포기 · **Front Immunol** 또는 **HLA** (IF 3–4) 직행
FRAME	Pan-Asian forest meta 만으로 frame · Bundang demote
COVER LETTER	HLA fallback version
PROBABILITY	~25%

FRONTIERS / MDPI 회피 원칙

Memory `v17_graves_pivot.md` 의 Frontiers 회피 원칙 과 Front Immunol fallback 채택 충돌 — **Front Immunol 은 fallback 용으로만 허용**. Default 는 J Autoimmunity. MDPI 는 어떤 시나리오에서도 회피.

§ 7

Cross-paper boundary — symmetric reference

Paper 2 ↔ Paper 3 두 references 가 symmetric 해야 reviewer 의 "왜 한 paper 로 안 묶었나" Q 사전 차단.

Paper 2 Discussion 한 줄

> *"Korean PTC HLA susceptibility (this paper) shares background with Korean Graves' (Paper 3, Cook et al. in prep), but PTC ≠ Graves' as diseases — PTC = neoplasia, Graves' = autoimmune hyperthyroidism, autoimmune-overlap PTC = autoimmune destructive infiltration. Pan-Asian thyroid HLA axis is shared, disease mechanisms distinct."*

Paper 3 Discussion 한 줄 (reciprocal)

> *"The Pan-Asian HLA susceptibility axis identified here for Graves' disease shares background alleles with Korean PTC (Paper 2, Cook et al. in prep) and autoimmune-overlap PTC, but mechanism diverges — Graves' = TSAb-driven hyperthyroidism, autoimmune-overlap PTC = destructive infiltration, PTC = thyroid neoplasia. Shared HLA risk reflects population-level susceptibility; disease specificity is determined downstream."*

두 references 의 **symmetric structure** 가 핵심 — reviewer 가 "한 paper 로 합칠 수 있지 않나?" 묻는 시점에 두 paper 의 Discussion 이 동일한 mechanism-divergence framework 으로 답변 가능. 합치지 않은 이유는 **"shared HLA risk reflects population-level susceptibility; disease specificity is determined downstream"** — 이것이 trajectory 정당화의 핵심 phrase.

§ 8

Timeline · post-entry 4–6 weeks

Entry conditions 4/4 충족 후 Tasks 1–4 진행.

2026-05-04 *GATED — 진입 보류 시작*

Memory `v19_paper3_GD_gating.md` 작성 + advisor isolation directive · Source file touch 금지.

2026-05-04 → 2026-06-13 *Wait — Paper 1 marathon W1–W6 진행 중*

Paper 1 bioRxiv 제출 + Paper 2 Task A/B/C 완료 대기. Paper 3 file 미접근.

2026-06-15 (조건부) *Entry conditions 4/4 충족 시 진입*

Yu professor Paper 3 trajectory 미팅 + Bundang 응답 결과 통합 결정.

W1 (2026-06-15 → 2026-06-21) *Task 1 — Bundang Graves' arcasHLA*

RNA-seq fastq → arcasHLA imputation · Azure burst $5–10 estimated.

W2 (2026-06-22 → 2026-06-28) *Task 2 — Pan-Asian forest meta*

Korean GD + Chu 2018 Chinese GD · DerSimonian-Laird random-effects · 6 alleles direction-consistency check.

W3 (2026-06-29 → 2026-07-05) *Task 3 — cookHLA SNP cross-platform validation*

SNP panel (협상 시) · 4-digit imputation · RNA-seq imputation 과 cross-validation.

W4 (2026-07-06 → 2026-07-12) *Task 4 — Outline (본인 voice 키보드)*

venue commit (Bundang n + effect size 기반) · Yu 미팅 안건 · 5-pillar Outline.

W5–W6 (2026-07-13 → 2026-07-26) *Manuscript draft*

Methods + Results + Discussion · 본인 voice 영역 (Hook · Discussion 첫 paragraph · Limitations) 본인 키보드.

§ 9

Discussion questions for advisor

Paper 1 + Paper 2 완료 직후 미팅 안건.

#	QUESTION	DECISION NEEDED BY
Q1	"Bundang prospective Graves' cohort outreach 결과 — 응답 vs 무응답 freeze 어떻게?"	분당 응답 또는 6주 freeze 시점
Q2	"venue ladder Scenario A/B/C — n 확정 후 어느 시나리오 commit?"	Task 2 완료 직후
Q3	"cookHLA SNP cross-platform validation — Bundang SNP panel 협상 가능?"	Bundang 응답 시
Q4	"Pillar V GD-specific haplotype dissection — Paper 2 와 cross-reference 어디까지?"	Outline 작성 직전
Q5	"Frontiers/MDPI 회피 원칙 vs Front Immunol fallback — 충돌 해소?"	Yu 미팅

Colophon

Set in Cormorant Garamond + Newsreader + JetBrains Mono. Burgundy / Gold / Sage palette. Single HTML · CSS inline.

Source & provenance

Memory · `v19_paper3_GD_gating.md` + `v17_graves_pivot.md`. Source files · 진입 후 generation (현재 미존재). Reference cohort · Chu X et al. 2018 J Med Genet 55(10):685–692.

Scope & boundaries

Paper 3 · Korean Graves' HLA + Pan-Asian autoimmune-thyroid landscape.
Paper 1 · 2 territory terms · 0 plain-text occurrences. 본인 cookHLA Nat Commun unique angle.